

Solar Portfolio Assessment

10 candidate sites

1	Lost Hills Ag 3	89	High	HIGH
2	Rosamond Corridor B (near Valentine)	87	High	HIGH
3	Rosamond Corridor D (near Willow Spring)	87	High	HIGH
4	Rosamond Corridor E (Solar Star core)	86	High	HIGH
5	Buttonwillow Ag 1	86	High	HIGH
6	South Kern Ag 5	85	High	HIGH
7	Rosamond Corridor C (near Catalina)	83	High	HIGH
8	Rosamond Corridor A (near Golden Field)	78	High	HIGH
9	Wasco Ag 2	Excluded	Excluded	HIGH
10	Shafter Ag 4	Excluded	Excluded	HIGH

Lost Hills Ag 3

Solar Site Suitability Assessment

LOCATION: Lost Hills Ag 3 · NERC REGION: WECC

SITE SCORE

89 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: Highway 46 Substation (3.4 mi, 115 kV)

Existing capacity (within 50 km): 639.1 MW across 53 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	97	hifld_transmission	HIGH
solar_ghi	0.10	62	nrel_pvwatts_v8	HIGH
solar_slope	0.17	99	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	90	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Rosamond Corridor B (near Valentine)

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor B (near Valentine) · NERC REGION: WECC

SITE SCORE

87 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (1.8 mi, 220 kV)

Existing capacity (within 50 km): 4501.3 MW across 138 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	98	hifld_transmission	HIGH
solar_ghi	0.10	96	nrel_pvwatts_v8	HIGH
solar_slope	0.17	81	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	94	usgs_3dep	HIGH

solar_land_cover	0.08	60	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Rosamond Corridor D (near Willow Spring)

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor D (near Willow Spring) · NERC REGION: WECC

SITE SCORE

87 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (0.9 mi, 220 kV)

Existing capacity (within 50 km): 4530.8 MW across 145 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	100	hifld_transmission	HIGH
solar_ghi	0.10	92	nrel_pvwatts_v8	HIGH
solar_slope	0.17	94	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	60	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	35	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Rosamond Corridor E (Solar Star core)

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor E (Solar Star core) · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.1 mi, 220 kV)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	91	hifld_transmission	HIGH
solar_ghi	0.10	93	nrel_pvwatts_v8	HIGH
solar_slope	0.17	95	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	60	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Buttonwillow Ag 1

Solar Site Suitability Assessment

LOCATION: Buttonwillow Ag 1 · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.6 mi)

Existing capacity (within 50 km): 464.0 MW across 47 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	92	hifld_transmission	HIGH
solar_ghi	0.10	67	nrel_pvwatts_v8	HIGH
solar_slope	0.17	99	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	70	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

South Kern Ag 5

Solar Site Suitability Assessment

LOCATION: South Kern Ag 5 · NERC REGION: WECC

SITE SCORE

85 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (4.1 mi)

Existing capacity (within 50 km): 427.6 MW across 42 operating plants (EIA Form 860).

Queue activity (CAISO): 52 active solar interconnection requests totaling 13169.4 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	86	hifld_transmission	HIGH
solar_ghi	0.10	65	nrel_pvwatts_v8	HIGH
solar_slope	0.17	99	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	90	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Rosamond Corridor C (near Catalina)

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor C (near Catalina) · NERC REGION: WECC

SITE SCORE

83 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (1.0 mi, 220 kV)

Existing capacity (within 50 km): 4902.4 MW across 143 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	99	hifld_transmission	HIGH
solar_ghi	0.10	96	nrel_pvwatts_v8	HIGH
solar_slope	0.17	72	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	96	usgs_3dep	HIGH

solar_land_cover	0.08	60	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	35	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Rosamond Corridor A (near Golden Field)

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor A (near Golden Field) · NERC REGION: WECC

SITE SCORE

78 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (0.6 mi, 220 kV)

Existing capacity (within 50 km): 4530.8 MW across 145 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Exclusion Screening Results		
Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

Scored Criteria Details				
solar_transmission	0.42	80	hifld_transmission	HIGH
solar_ghi	0.10	92	nrel_pvwatts_v8	HIGH
solar_slope	0.17	93	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	50	usgs_3dep	HIGH

solar_land_cover	0.08	60	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Wasco Ag 2

Solar Site Suitability Assessment

LOCATION: Wasco Ag 2 · NERC REGION: WECC

SITE SCORE

80 / 100

Excluded

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: Wasco Substation (1.1 mi, 115 kV)

Existing capacity (within 50 km): 641.4 MW across 56 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✗ EXCLUDED	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	92	hifld_transmission	HIGH
solar_ghi	0.10	65	nrel_pvwatts_v8	HIGH
solar_slope	0.17	97	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	0	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Shafter Ag 4

Solar Site Suitability Assessment

LOCATION: Shafter Ag 4 · NERC REGION: WECC

SITE SCORE

72 / 100

Excluded

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.8 mi)

Existing capacity (within 50 km): 735.9 MW across 56 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✗ EXCLUDED	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	73	hifld_transmission	HIGH
solar_ghi	0.10	66	nrel_pvwatts_v8	HIGH
solar_slope	0.17	99	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH

solar_land_cover	0.08	0	nlcd_land_cover	HIGH
solar_soil	0.05	100	usda_sda_ssurg	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations